LAPORAN WORKSHOP DEVELOPER OPERATIONAL

MINGGU 12

Oleh:

Moh. Farhan Assidiqi (E41230806)

PROGRAM STUDI TEKNIK INFORMATIKA

JURUSAN TEKNOLOGI INFORMASI

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PRAKTEK DEPLOY DAN PENERAPAN REPLICA WEB PADA KUBERNETES

1. Menyiapkan folder dan membuat proyek Laravel

```
C:\Users\ADVAN\Documents
λ cd kubernetes-laravel

C:\Users\ADVAN\Documents\kubernetes-laravel
λ ls
laravel-minikube/

C:\Users\ADVAN\Documents\kubernetes-laravel
λ cd laravel-minikube

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ ls

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ cd ..

C:\Users\ADVAN\Documents\kubernetes-laravel
λ rm laravel-minikube
rm: cannot remove 'laravel-minikube': Is a directory

C:\Users\ADVAN\Documents\kubernetes-laravel
λ rm -rf laravel-minikube

C:\Users\ADVAN\Documents\kubernetes-laravel
λ composer create-project laravel/laravel laravel-minikube
```

2. Copy env dan lakukan key generate

```
C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ cp .env.example .env

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ php artisan key:generate

  INFO  Application key set successfully.
```

3. Start minikube dan pengecekan

```
C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Creating docker container (CPUs=2, Memory=3900MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   30s   v1.35.1
```

4. Konfigurasi Dockerfile

```
FROM php:8.3-apache

RUN docker-php-ext-install pdo pdo_mysql
RUN a2enmod rewrite

ENV APACHE_DOCUMENT_ROOT=/var/www/html/public

RUN sed -ri -e 's!/var/www/html!${APACHE_DOCUMENT_ROOT}!g' /etc/a
RUN sed -ri -e 's!/var/www/html!${APACHE_DOCUMENT_ROOT}!g' /etc/a

COPY --chown=www-data:www-data . /var/www/html

RUN chmod -R 775 /var/www/html/storage /var/www/html/bootstrap/ca

EXPOSE 80
```

5. Konfigurasi file Laravel.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: laravel
spec:
  replicas: 3   # scaling
  selector:
    matchLabels:
      app: laravel
  template:
    metadata:
      labels:
        app: laravel
    spec:
      containers:
      - name: laravel
        image: laravel-app
        imagePullPolicy: IfNotPresent
        env:
        - name: DB_HOST
          value: mysql
        - name: DB_DATABASE
          value: laravel
        - name: DB_USERNAME
          value: root
        - name: DB_PASSWORD
          value: root
        ports:
        - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: laravel
spec:
  selector:
```

6. Konfigurasi mysql.yaml

```yaml
metadata:
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
      - name: mysql
        image: mysql:8
        env:
        - name: MYSQL_ROOT_PASSWORD
          value: root
        - name: MYSQL_DATABASE
          value: laravel
        ports:
        - containerPort: 3306
---
apiVersion: v1
kind: Service
metadata:
  name: mysql
spec:
  selector:
    app: mysql
  ports:
    - port: 3306
```

7. Konfigurasi service.yaml

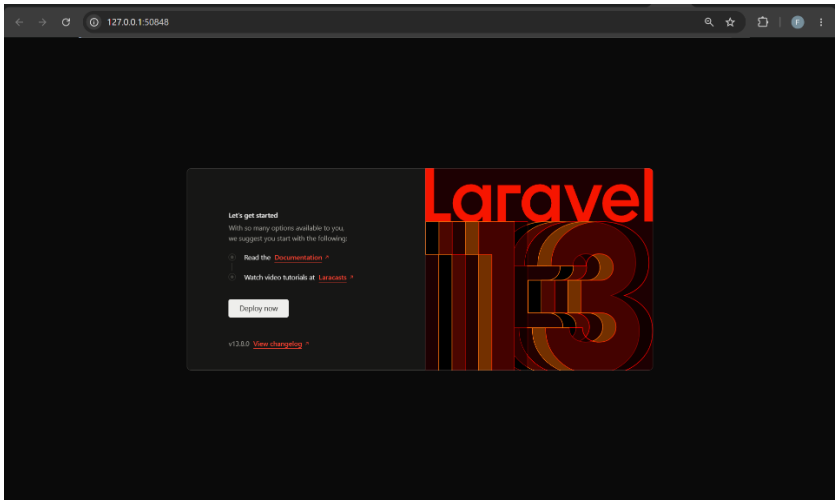
```yaml
apiVersion: v1
kind: Service
metadata:
  name: laravel-nodeport
spec:
  type: NodePort
  selector:
    app: laravel
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```

8. Deploy dan pengecekan

```
C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube\k8s
λ kubectl apply -f .
deployment.apps/laravel created
service/laravel created
deployment.apps/mysql created
service/mysql created
service/laravel-nodeport created

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube\k8s
λ kubectl get pods
NAME                        READY   STATUS              RESTARTS   AGE
laravel-575b8c55bf-2f4t7    0/1     ContainerCreating   0          4s
laravel-575b8c55bf-kc4pn    0/1     ContainerCreating   0          4s
laravel-575b8c55bf-qs7db    0/1     ContainerCreating   0          4s
mysql-5c4d589d75-vjxfq      0/1     ContainerCreating   0          4s

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube\k8s
λ kubectl get svc
NAME                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
kubernetes          ClusterIP   10.96.0.1        <none>        443/TCP        2m57s
laravel             ClusterIP   10.111.108.219   <none>        80/TCP         12s
laravel-nodeport    NodePort    10.109.175.238   <none>        80:30007/TCP   12s
mysql               ClusterIP   10.105.47.150    <none>        3306/TCP       12s
```

9. Build dan Jalankan aplikasi

```
λ docker build -t laravel-app .
[+] Building 2.7s (13/13) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                   0.0s
 => => transferring dockerfile: 501B                                                                   0.0s
 => [internal] load metadata for docker.io/library/php:8.3-apache                                      1.7s
 => [auth] library/php:pull token for registry-1.docker.io                                             0.0s
 => [internal] load .dockerignore                                                                      0.0s
 => => transferring context: 2B                                                                        0.0s
 => [internal] load build context                                                                      0.5s
 => => transferring context: 789.73kB                                                                  0.5s
 => [1/7] FROM docker.io/library/php:8.3-apache@sha256:3f2b61d568d10b04406b6128ad053197689eb13efeb5e1c499d7fd5a469a1219   0.0s
 => => resolve docker.io/library/php:8.3-apache@sha256:3f2b61d568d10b04406b6128ad053197689eb13efeb5e1c499d7fd5a469a1219   0.0s
 => CACHED [2/7] RUN docker-php-ext-install pdo pdo_mysql                                              0.0s
 => CACHED [3/7] RUN a2enmod rewrite                                                                   0.0s
 => CACHED [4/7] RUN sed -ri -e 's!/var/www/html!${APACHE_DOCUMENT_ROOT}!g' /etc/apache2/sites-available/000-default.conf   0.0s
 => CACHED [5/7] RUN sed -ri -e 's!/var/www/html!${APACHE_DOCUMENT_ROOT}!g' /etc/apache2/apache2.conf   0.0s
 => CACHED [6/7] COPY --chown=www-data:www-data . /var/www/html                                        0.0s
 => CACHED [7/7] RUN chmod -R 775 /var/www/html/storage /var/www/html/bootstrap/cache                  0.0s
 => exporting to image                                                                                 0.2s
 => => exporting layers                                                                                0.0s
 => => exporting manifest sha256:4da26b1e220249715a6659d6c22a75e384e110d8076fef6b33cbebc97ae59cd2      0.0s
 => => exporting config sha256:374070465920fa4feb0d8627a34a995fbdb225eef6585570570a364d72480b7e        0.0s
 => => exporting attestation manifest sha256:41ddc21f18e8c52abd790b7f386db826b584c33ffbd12764ce7cdefc65671f21   0.0s
 => => exporting manifest list sha256:0a3709a65b436da3cfc3c543d03c9d8029f3dce691a8c17323aad8516ff8054f   0.0s
 => => naming to docker.io/library/laravel-app:latest                                                  0.0s
 => => unpacking to docker.io/library/laravel-app:latest                                               0.0s

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ  minikube image load laravel-app

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ minikube service laravel-nodeport
|-----------|----------------|-------------|---------------------------|
| NAMESPACE |      NAME      | TARGET PORT |           URL             |
|-----------|----------------|-------------|---------------------------|
| default   | laravel-nodeport |  80       | http://192.168.49.2:30007 |
|-----------|----------------|-------------|---------------------------|
* Starting tunnel for service laravel-nodeport.
```



10. Melakukan migrasi db

```
C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ kubectl exec -it laravel-575b8c55bf-2f4t7 -- php artisan migrate

  INFO  Preparing database.

  Creating migration table ..................................................... 49.37ms DONE

  INFO  Running migrations.

  0001_01_01_000000_create_users_table ......................................... 189.31ms DONE
  0001_01_01_000001_create_cache_table ......................................... 125.33ms DONE
  0001_01_01_000002_create_jobs_table .......................................... 155.01ms DONE
```

11. Melakukan replica web

```
C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ kubectl scale deployment laravel --replicas=5
deployment.apps/laravel scaled

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ kubectl scale deployment laravel --replicas=2
deployment.apps/laravel scaled

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ kubectl get pods -w
NAME                       READY   STATUS    RESTARTS   AGE
laravel-575b8c55bf-kc4pn   1/1     Running   0          10m
laravel-575b8c55bf-qs7db   1/1     Running   0          10m
mysql-5c4d589d75-vjxfq     1/1     Running   0          10m
```

12. Clear environment

```
C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ kubectl delete deployment laravel
deployment.apps "laravel" deleted from default namespace

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ docker rmi laravel-app
Untagged: laravel-app:latest
Deleted: sha256:0a3709a65b436da3cfc3c543d09c9d8029f3dce691e8c17323aad8516ff8054f

C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ minikube ssh
Linux minikube 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
docker@minikube:~$ docker images
```

```
C:\Users\ADVAN\Documents\kubernetes-laravel\laravel-minikube
λ minikube delete --all --purge
* Deleting "minikube" in docker ...
* Removing C:\Users\ADVAN\.minikube\machines\minikube ...
* Removed all traces of the "minikube" cluster.
* Successfully deleted all profiles
* Successfully purged minikube directory located at - [C:\Users\ADVAN\.minikube]
* Kicbase images have not been deleted. To delete images run:
  - docker rmi gcr.io/k8s-minikube/kicbase:v0.0.50 gcr.io/k8s-minikube/kicbase:<none>
```